Forecasting Exchange Rates using Time Series Analysis

Objective

To forecast USD to AUD exchange rates using **ARIMA** and **Exponential Smoothing** models and compare their performance.

Part 1: Data Preparation and Exploration

Data Loading

- Dataset `exchange_rate.csv` loaded using pandas
- `date` column converted to datetime and set as index

Initial Exploration

- Time series plot shows:
    - Long-term fluctuations
    - No strong seasonality
    - Presence of trend
- Rolling mean used to analyze trend stability

Data Preprocessing

- Checked for missing values → handled using forward fill
- Outliers smoothed using rolling mean
- Stationarity checked using ADF Test

Part 2: ARIMA Model

Parameter Selection

- ACF and PACF plots used:

- ACF → helps choose **q**
- PACF → helps choose **p**
- Differencing applied → to make data stationary (**d = 1**)

Model Fitting

- ARIMA model trained on training dataset

Diagnostics

- Residual plots show no pattern
- Residuals follow near normal distribution

Forecasting

- Forecast performed on test dataset
- Predictions plotted against actual values

Part 3: Exponential Smoothing

Model Selection

- Holt's Linear Trend model selected (trend present, no seasonality)

Parameter Optimization

- AIC-based comparison used
- Best smoothing parameters selected automatically

Forecasting

- Model fitted and predictions generated
- Compared visually with actual data

Part 4: Evaluation and Comparison

Metrics Used

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- MAPE (Mean Absolute Percentage Error)

Model Comparison

Model	Performance
ARIMA	Good for capturing patterns but sensitive to parameters
Exponential Smoothing	Better for trend-following data

Observations

- ARIMA captures statistical dependencies
- Exponential Smoothing reacts better to recent trends
- Exchange rate volatility affects both models

Conclusion

Exponential Smoothing performed slightly better due to its ability to adapt to recent trends. However, ARIMA remains powerful for structured time series modeling. A hybrid approach could further improve accuracy.